Project Porfolio

Zhao Zehui (Leila)
Mechanical Engineering (Year 5)
The University of Hong Kong



Github: https://github.com/Leilazehui/Leilazehui.github.io/tree/main

About Me

Interest directions:

- Aerospace Systems Engineering
- Space Systems
- Robotics & Autonomous Systems
- Design-space Optimzation

Interested industries:

- System Design and Product Development in
  - Automotive
  - Aviation
  - Large mechanical systems to tackle real-life challenges

About Me

Specialization:

- Aerospace Vehicle Design
- Rapid-Prototyping
- Design-Space Optimization

Skills:

- SolidWorks (since fall 2023),
- AutoCAD (since summer 2024)
- MATLAB/Simulink (since fall 2024),
- Xflr5 (since fall 2025),
- Ansys (since fall 2025),
- Arduino (since fall 2024),
- Laser-cutting machine, Workbench tools, lathe machine, drilling machine (since summer 2024)

About Me

Key achievement:

- DBF Team Leader and Chief Engineer, HKD 100,000 funding secured
- Top 32/98 worldwide, 3rd in Asia, 2nd in Hong Kong
- Founder of AIAA Student Branch - HKU
- Advanced Aerospace Coursework GPA: 3.52/4.3
- 4+ Engineering-related internship experiences (HKU SRA, HKPC, 2 Startups, Towngas)
- 4+ Engineering-related projects (AIAA DBF 2026, CANSAT Satellite 2026, RAICOM 2025, Soft robotics 2025, 3D printing 2024, 4 Course projects 2023 - 2026, Robomaster 2023,)
- Roles: Chief engineer, team leader, software lead, mechanical team member, student technician)

Project & Internship List

1. UAV Design and Drone Control	2026 - 2026
2. CANSAT Satellite Design - Software Lead	2026 - 2026
3. AIAA DBF Aircraft Design Competition 2026 - Team Leader/Chief Engineer	2025 - 2026
4. Trade-space framework for mission-driven aircraft design decision - First Author	2025 - 2026
5. HKPC Internship - Winter Intern in Smart Manufacturing Division	2025 - 2026
6. UAV Design Project - Motor and Battery Research	2025 - 2026
7. Building Service Project - Soil & Waste System 2D layout design (AutoCAD)	2025 - 2026
8. Power Transmission Unit System for Mining Industry - Team representative	2024 - 2025
9. WeGoAlgae Employee - Mechanical & Electrical Engineer	2024 - 2025

Project List

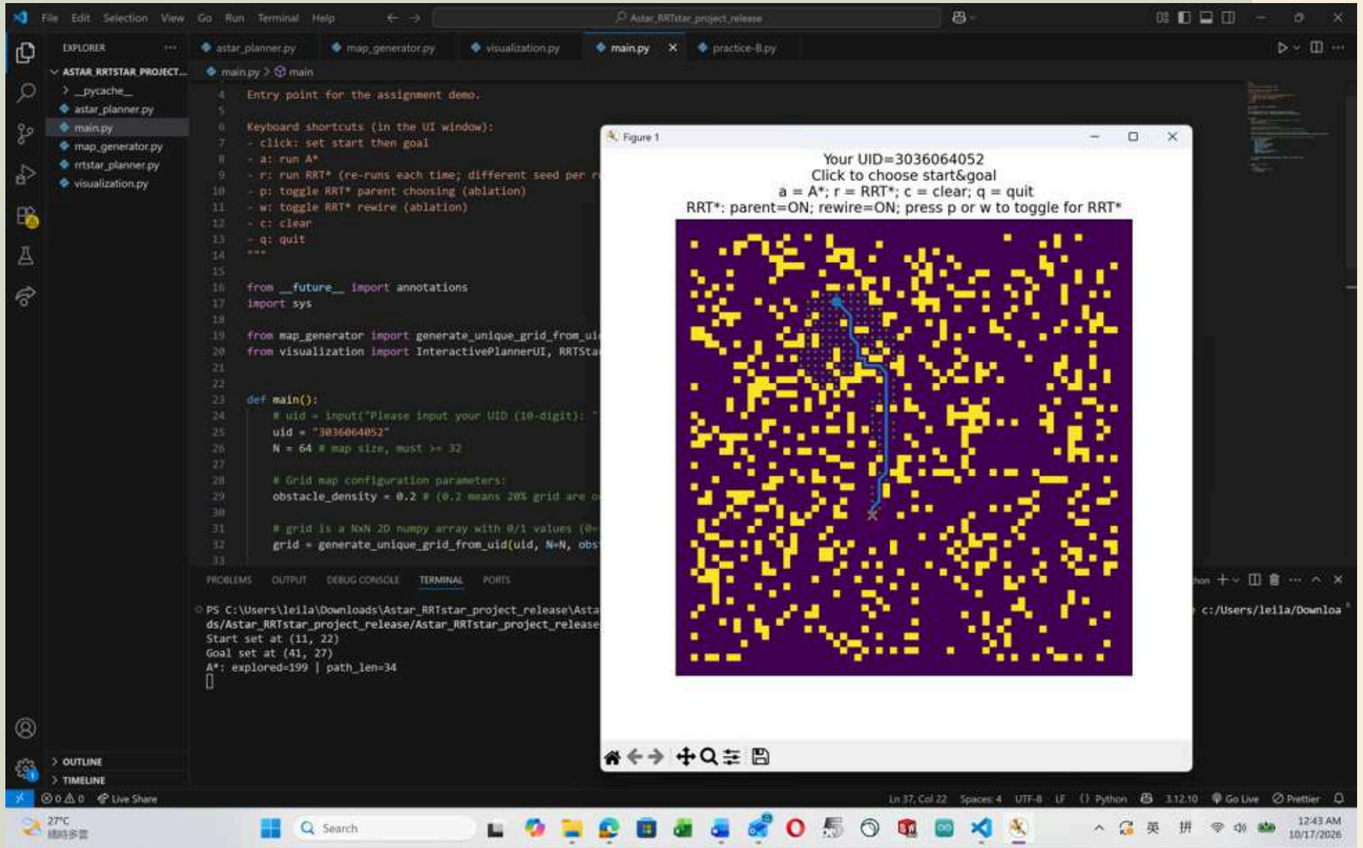
9. Soft Robotics Study 2024 - 2025
10. 3D Printing Project - 3D printing material study and structure design for weight reduction of wing ribs 2024 - 2025
11. HKU Summer Engineering Training 2023 - 2024
12. Advanced Mechanical Engineering Core Course Performance
13. Academic - Education Project - HKUDBF Team Leader 2024 - 2026

UAV Design and Drone Control

Sept 2026 - now

- Drone path planning algorithm study on Python (Window and Linux), including GFS, Dijkstra, GBFS, A* and RRT*. Architected and implementing A* and RRT* algorithm utilizing Visual Studio Code and Python.

- For code and more information, please visit my Github porfolio

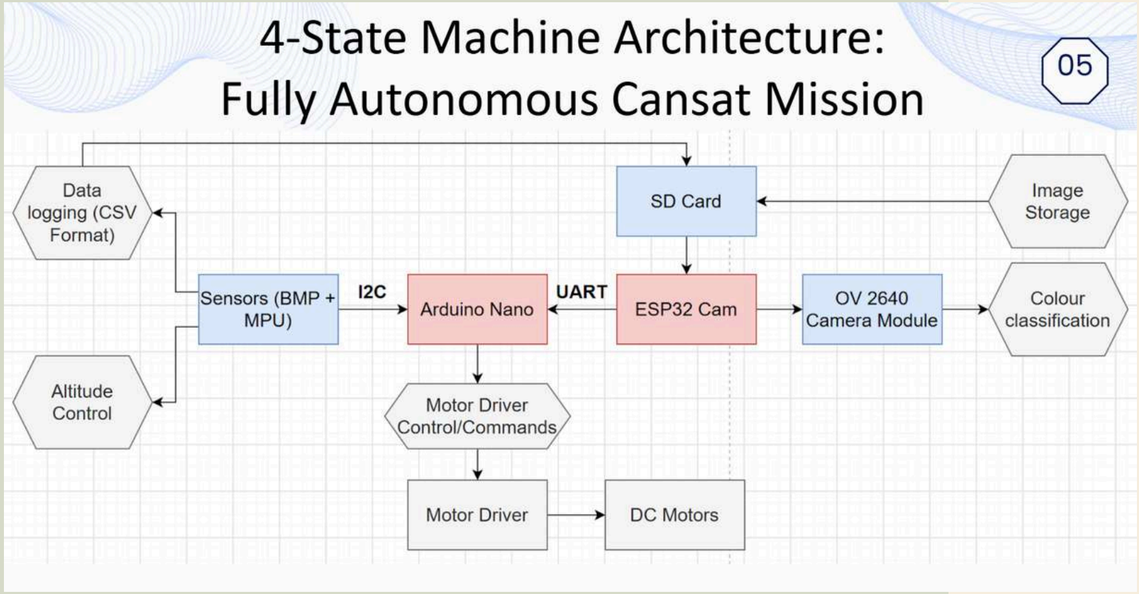


Jul - Aug 2026

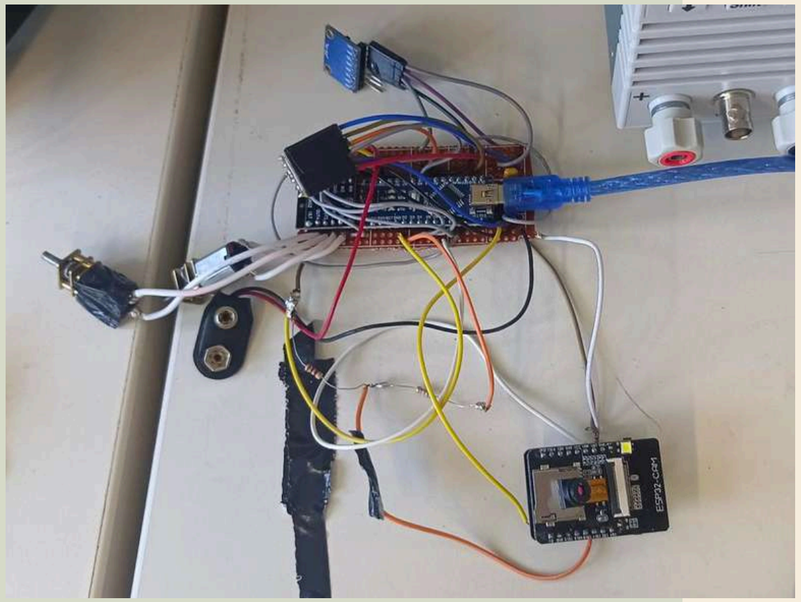
CANSAT Design

Software Lead & Embedded System Architect

- Flight computer and system architecture and testing
  - 4-state machine architecture breakdown for an autonomous landing and navigation CanSat mission
  - Sensors integration and colour classification mechanism testing
- Flight data analysis
  - Brief data analysis on satellite mission result







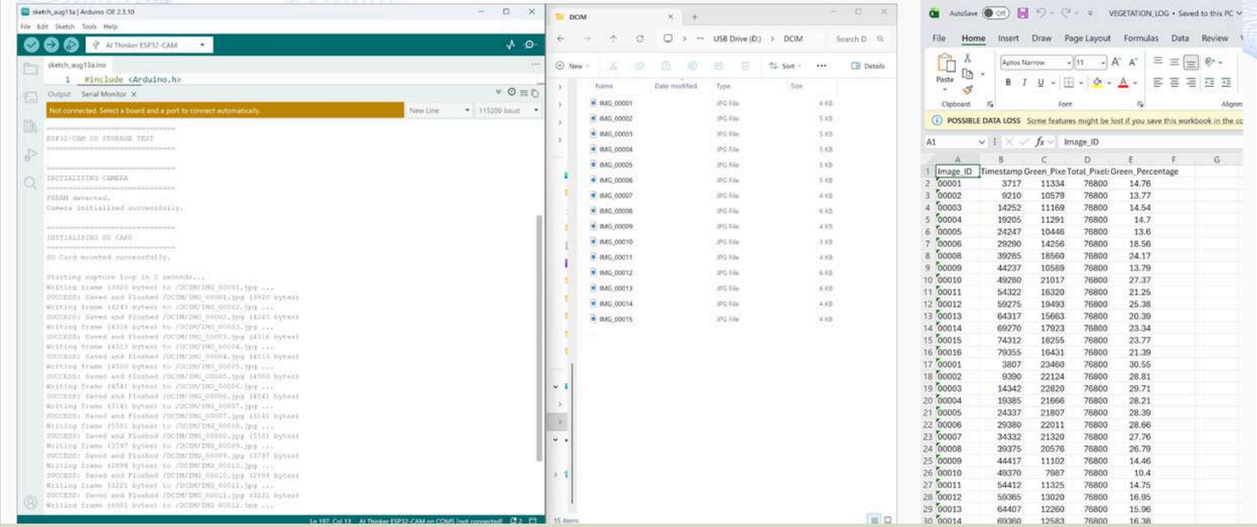
CANSAT Design

Software Lead & Embedded System Architect

Jul - Aug 2026

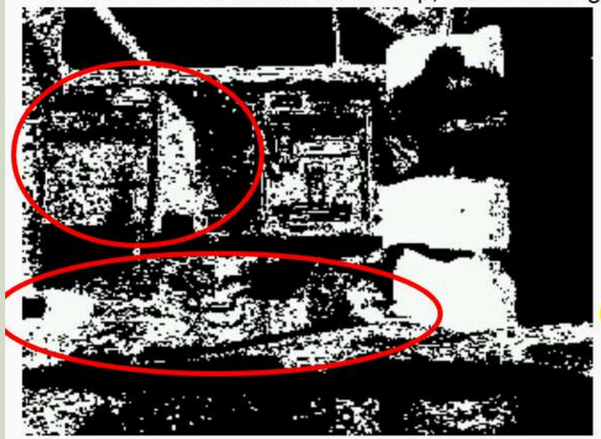
- Post-flight evaluation, testing, and Validation
  - Evaluate cansat performance, end-to-end object detection and avoidance mechanism testing
- Rapid prototyping through AI vibe coding techniques

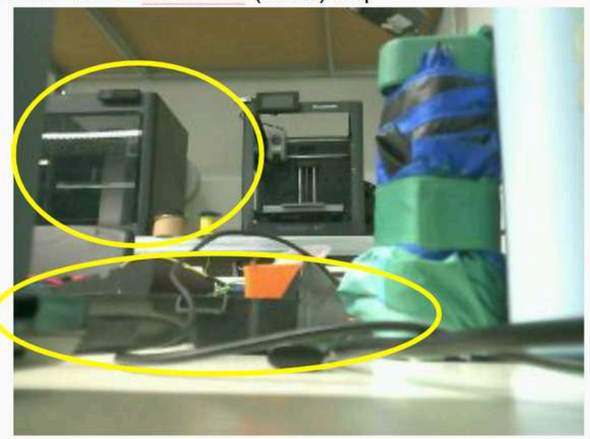
Verification: Real-time Image Storage in SD Card



Object Detection & Avoidance - Testing & Validation

- Version 3: Improved precision of colour classification
- Solved: RGB values inversion trap, Noise-filtering, Auto-White-balanced (AWB) trap



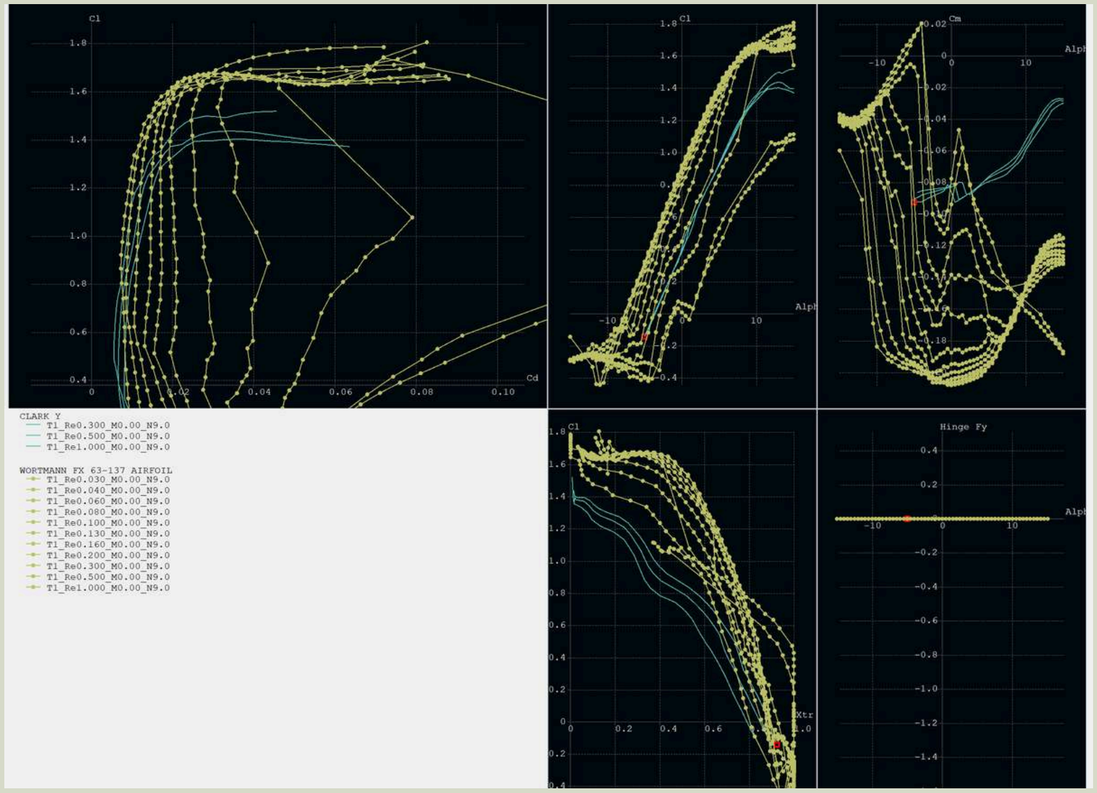


Full presentation available on github

Project AIAA: Aerodynamic Analysis

July 2025 - April 2026

- Airfoil aerodynamic analysis and comparison under different configurations
- Comparison on aerodynamic performance for main-wing on XFlr5



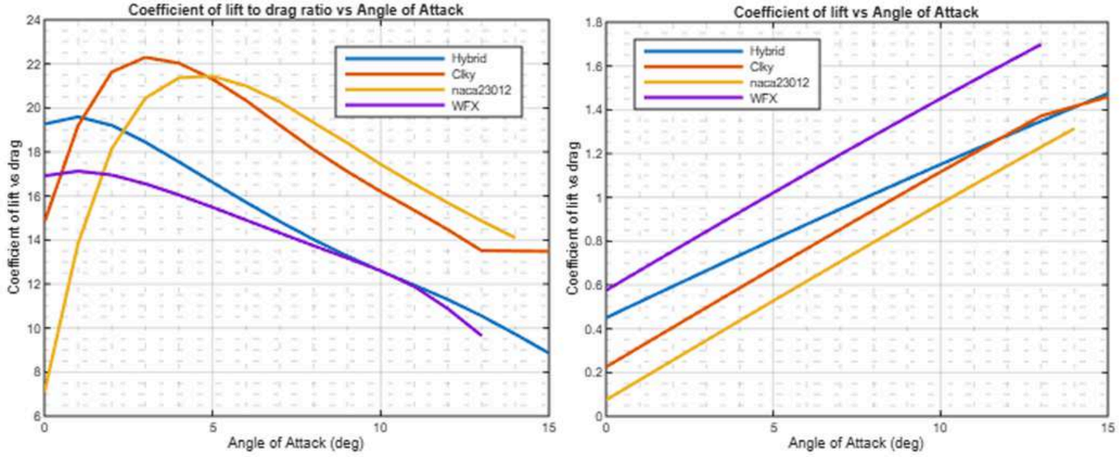


Figure 7 a) Coefficient of Lift-Drag Ratio vs Angle of Attack and b) Coefficient of Drag vs Angle of Attack

Characteristics	Weighting	Clark-Y	WFX-63 137	NACA23012	Hybrid
CL/CD for turn	30	1	0.5	0.75	0.25
CL/CD for cruise	25	0.5	0.75	0	1
Maximum Lift Coefficient	20	0.5	1	0.25	0.75
Manufacturability	25	1	0.25	0.25	0.5
	Total	67.5	55	33.75	60

Table 11 Comparison of Airfoil Performance

Project AIAA: Stability Analysis

- Analyze the lateral motion and longitudinal motion of the aircraft in XFLR5 simulation

- 5 aircraft modes data were recorded and converted into eigenvalues: decay of rate of the motion (σ in real axis) and the frequency of oscillation, (ω in the imaginary part)

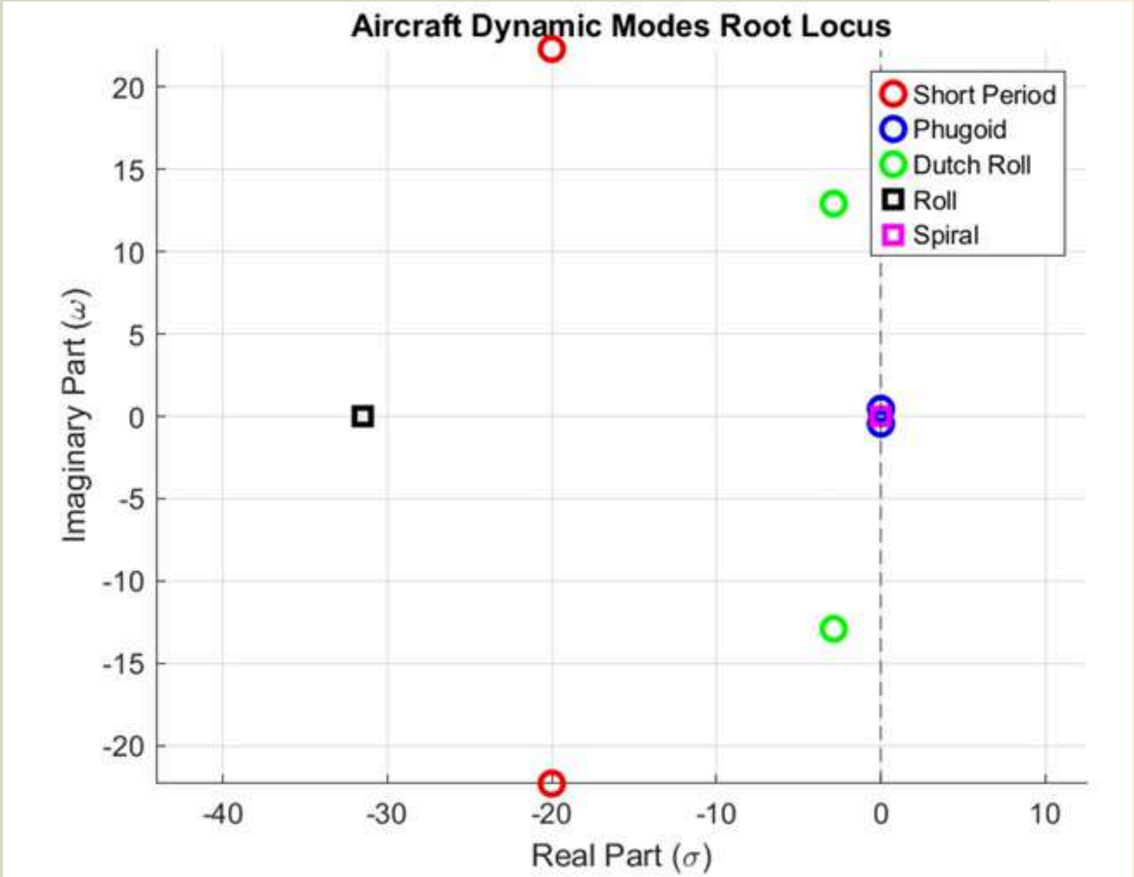
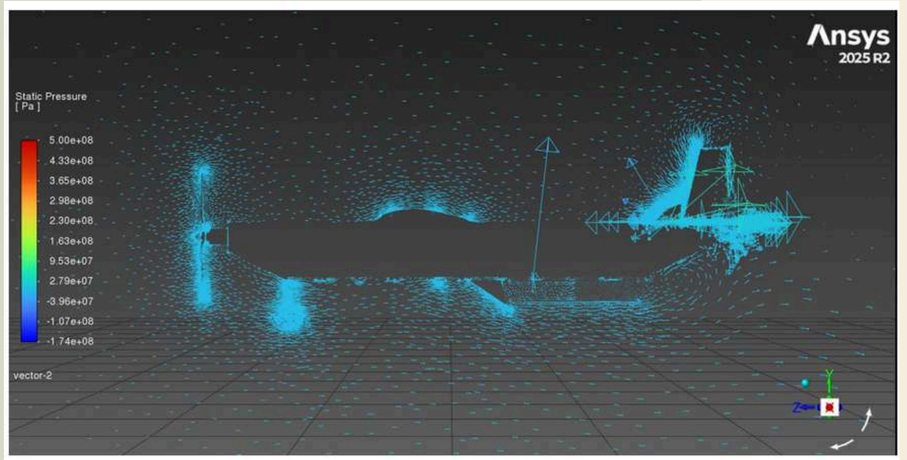


Figure 31 Aircraft Dynamic Modes Root Locus

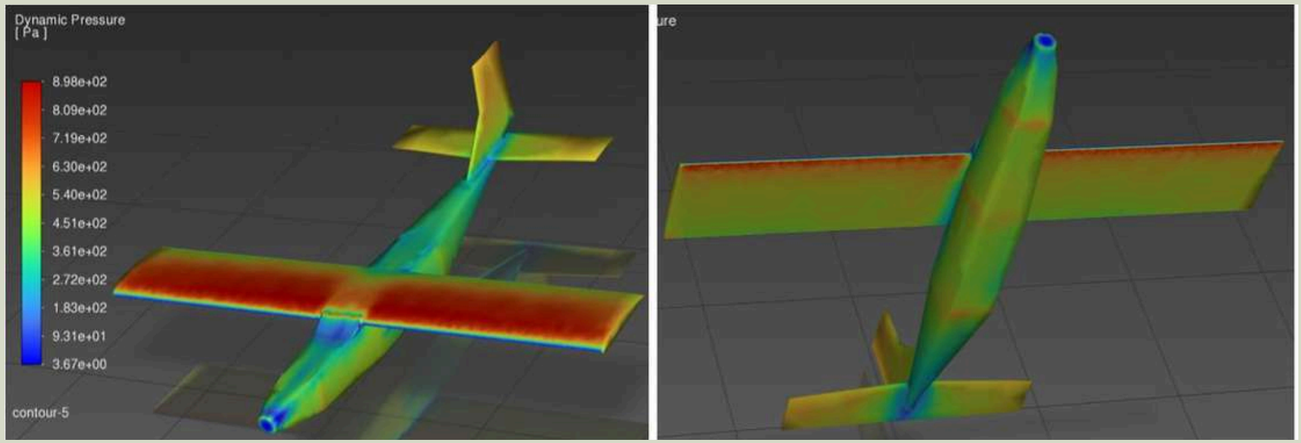
Project AIAA: CFD Analysis

- 3 prototypes with various configuration were simulated on ANSYS and analyzed to obtain static and dynamic pressure, velocity distribution and lift and drag.

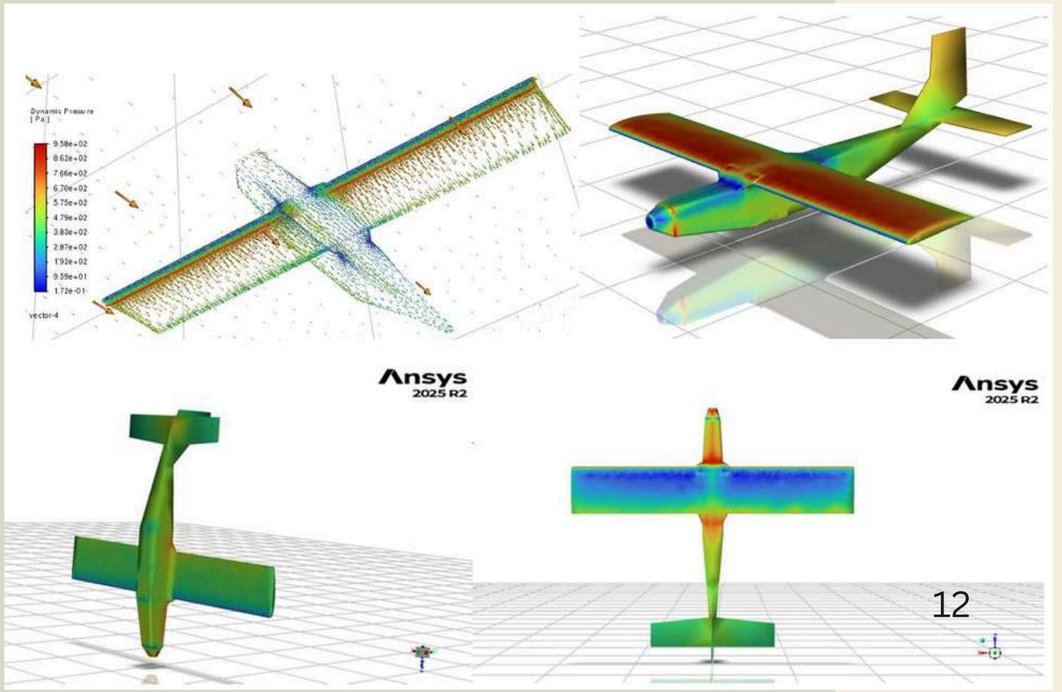
Prototype 1



Prototype 3



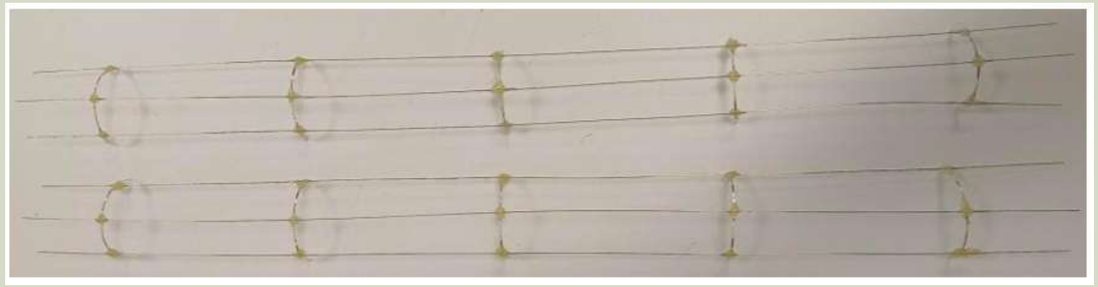
Prototype 2



Features	Prototype 2	Prototype 3
Side panel shape	Flat	Curved/round
Lift force (N)	41.5	34.72
Drag force (N)	6.5	7.15

Project AIAA: Banner Stow, tow and deploy mechanism & Drag test simulation

- 5 customed banner deploy mechanism and 2 release mechanism



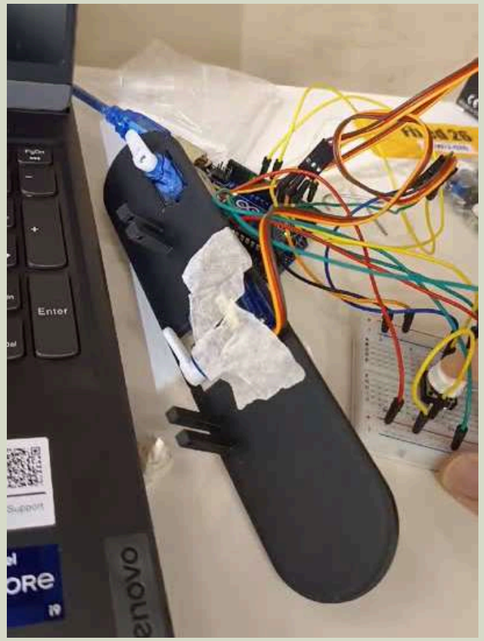
cage grabber



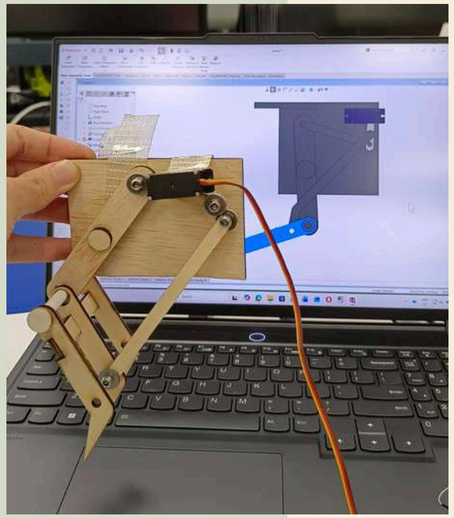
Rubber band



Pillars



Arduino-driven pillar mechanism



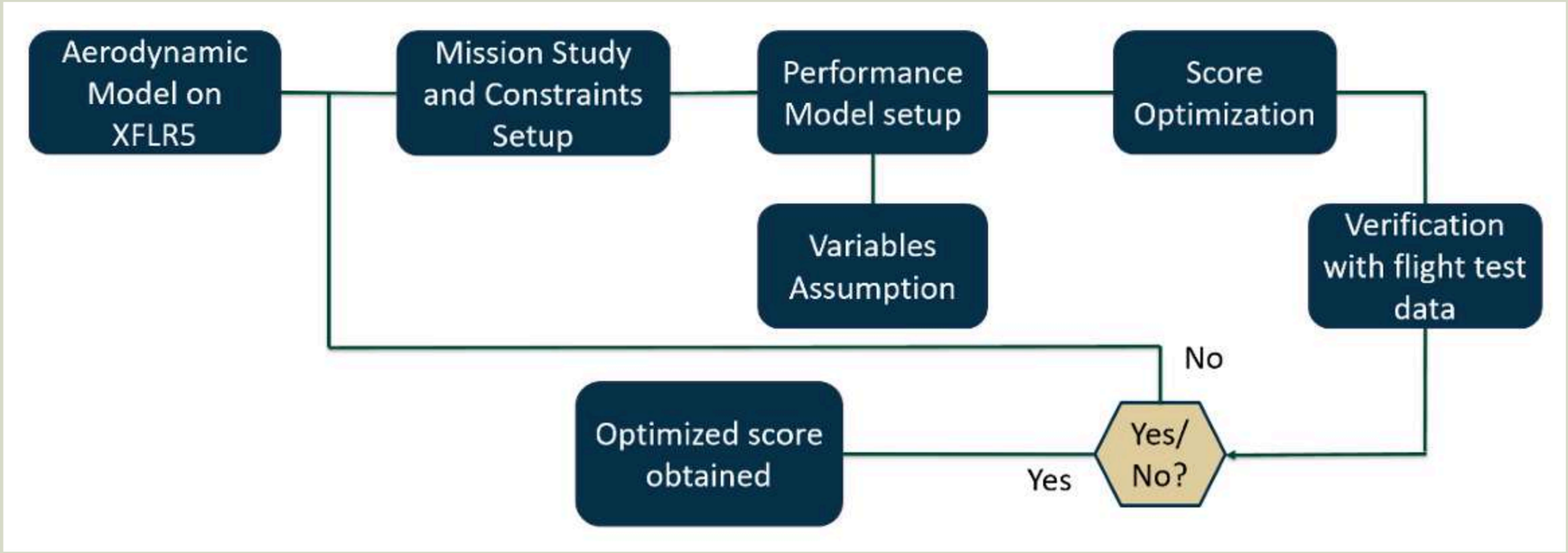
Bucket

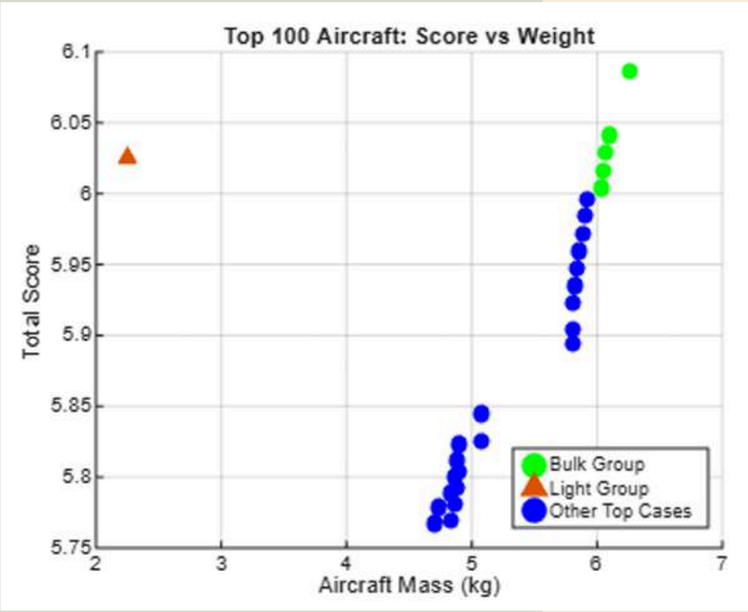


Gripper

Trade-Space Exploration and Decision Framework for Mission-Driven Aircraft Design

- Flowchart of the optimization framework
- An automated grid-search algorithm to simulate hundreds of cases of aircraft performance under different configurations on MATLAB and compare for 15 parameters



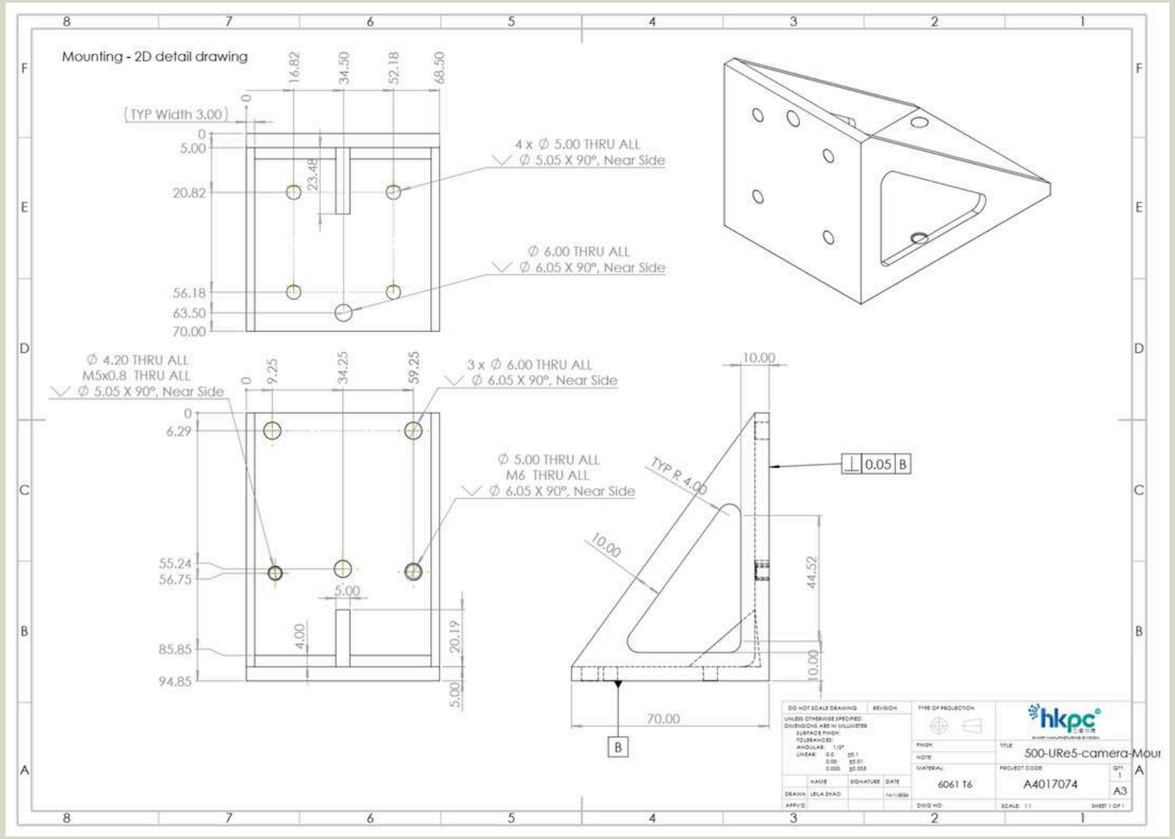


More explanation available on Github

Intern HKPC

Dec 2025 - Jan 2026

- 6-axes robotic arm and 3D camera Mounting Device Design on SolidWorks
- AI detection model on Anomaly and normal operational temperature of the robotic arm





Project UAV Design: Motor and Battery Study

MECH 4427	Innovative design and unmanned	6.00	Grading Basis 4.3	A+	25.80

Sept 2025 - Dec 2025

	Weight	KV	Max Thrust	Max Current	Peak Power	Voltage	Propeller Size	Temperature	Price in USD
Scorpion A-4220-540kv V2	288	540	6980g	115	2553	22.2	16*8	-	199.99
Sunnysky X4120 V3	292	480	6780g	97	1793.76	22.2	16*8	130	105.99
Dualsky XM5060 EA-10	392	490	8308g	110.2	2280	22.2	16*10	-	112.06
T-motor AT4125 Long Shaft	355	540	8492g	85	1718.7	22.2	16*8	97	115.99

	Weight	KV	Max Thrust	Max Current	Peak Power	Voltage	Propeller Size	Temperature	Price in USD
Scorpion A-3017	136g (154g mounted)	600	5030g	43	962	22.2	14*6	-	129.99
Sunnysky X3520 V3	219	560	4430g	80	1478.52	22.2	14*7	90 (1 min)	59.99
Dualsky XM4255 EA-12	269	520	5045g	59.2	1344	22.2	14*7	-	72.7
SunnySky X Series V3 X3120	159g	585	3560	65	1625	22.2	12*6.5	130 (3 min)	47.99

Worked on Avionics, particularly motor and battery trade-off study for RC aircraft to carry a 4-meter-long banner

AVIONICS

- **Servo**: KM0940MD
- **LiPo Battery**: 6-cells, 3300 mAh, 30C



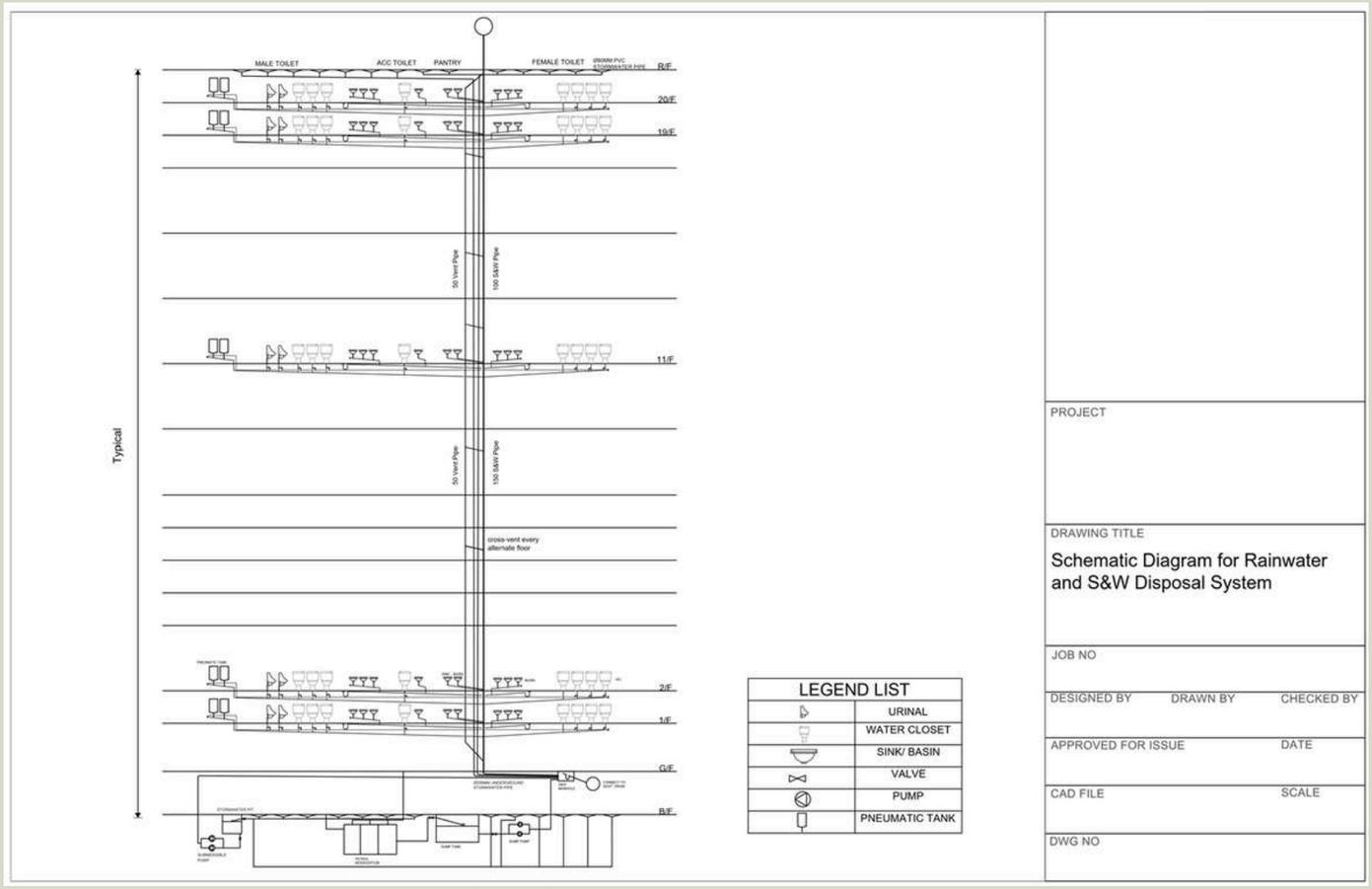


Prepared by Zhao Zehui

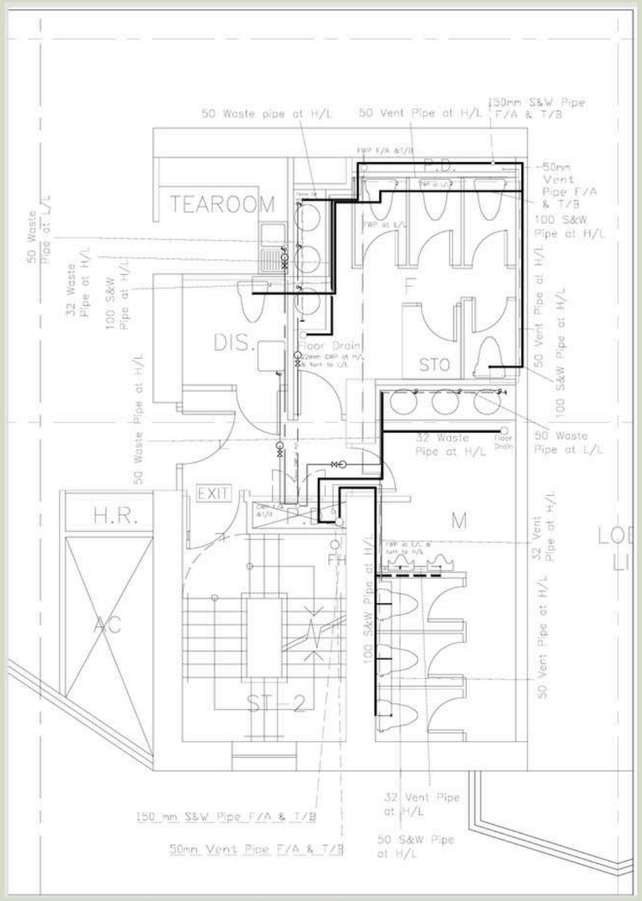
Project Building Service: Soil & Waste System

- Designed soil and waste (S&W) system for the plumping and drainage service of a factitious office building project collaborated with 2 students.

Sept - Dec 2025



S&W and rain water system layout



Toilet pipeline layout

MECH 3431	Utility services in buildings	6.00	Grading Basis 4.3	A-	22.20

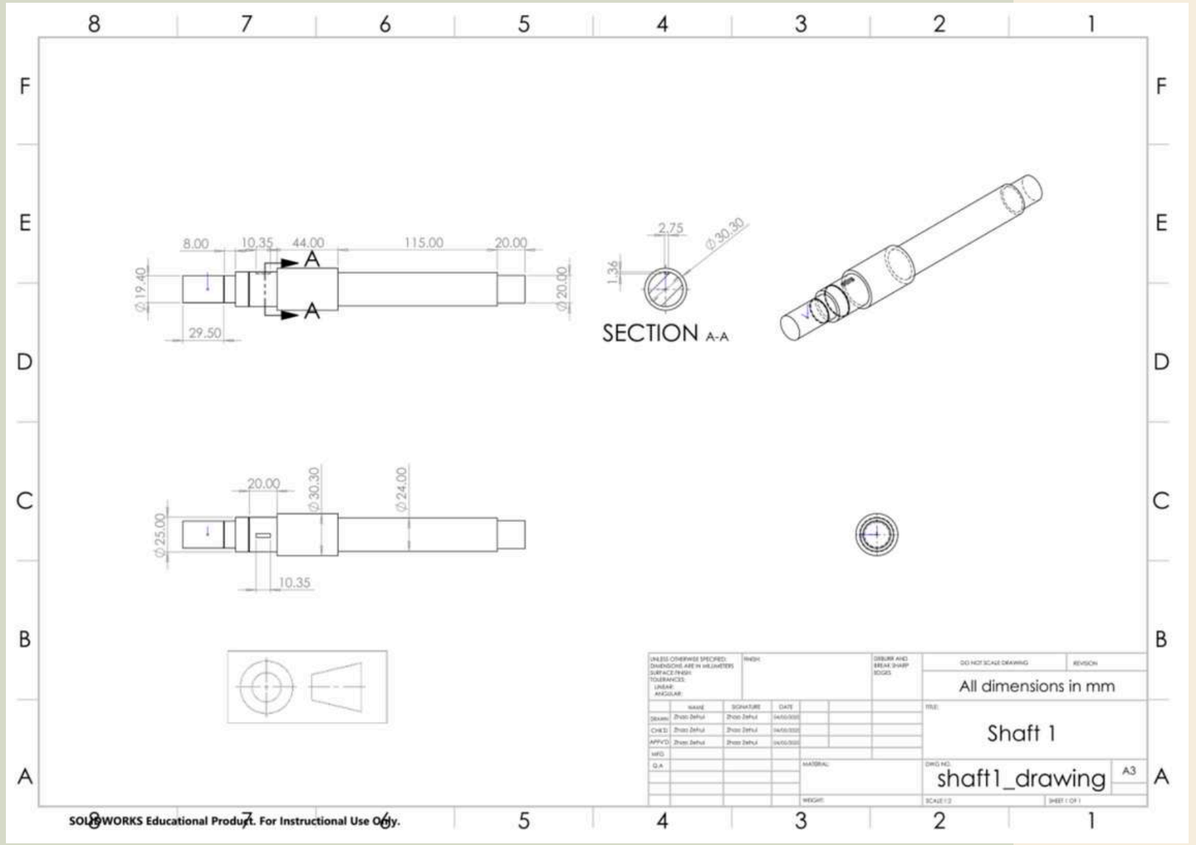
Project Engineering Design: Power transmission unit system

Jan 2025 - Apr 2025

- Designed a gear box for power transmission unit system mainly for mining industry
- Responsibilities:

Layout of the system
Chain drive calculation
Chain drive market research
Gear design & calculation
Shaft design & calculation
Shafts model & landscape drawing
Shaft market research
Bearing calculation
Key & Keyseat design & calculation
Key & Keyseat model & landscape drawing
Key & Keyseat market research

- Course result: A+



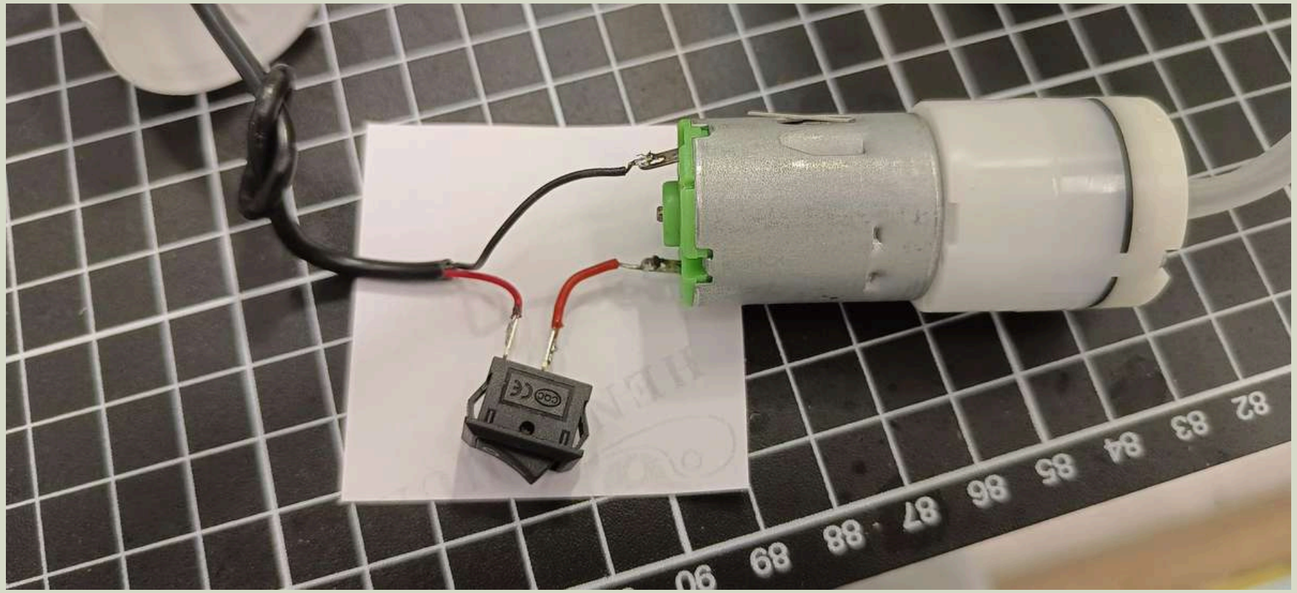
- Skills: Market research, SolidWorks model design, Engineering calculation

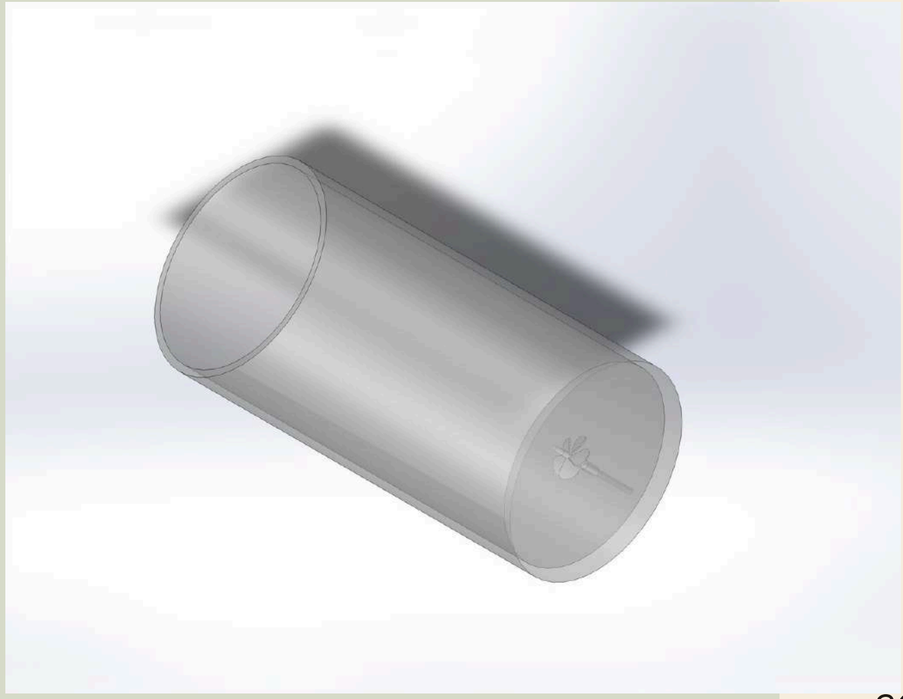
MECH 2404	Drawing and elements of design	2023-24 Sem 1	A+	6.00	✓
MECH 3427	Design and Manufacture	2024-25 Sem 2	A+	6.00	✓

Project WeGoAlgae

Jul 2024 - Jan 2025

- Designed and connected a circuit to integrate a 5V air pump, switch and a power adaptor
- Designed a conceptual cylinder water tank with water-proof turbine for improving water circulation rate and algae photosynthesis efficiency.





Soft Robotics Study

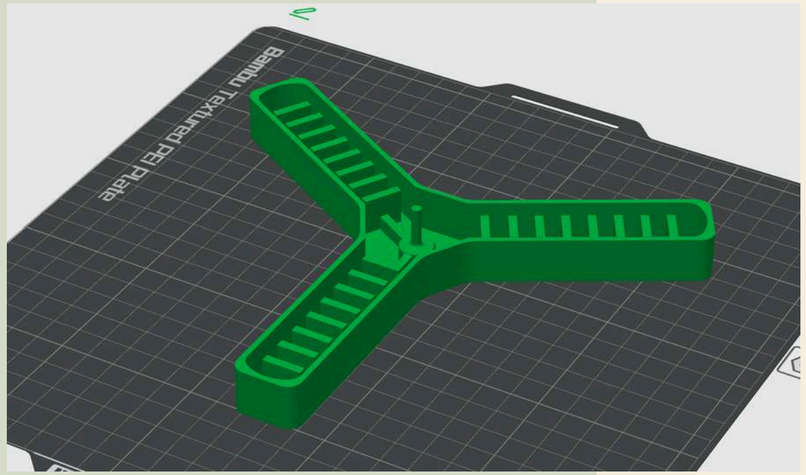
Jan - Jul 2025

- Study on locomotion of a pneumatic-driven octopus device fabricated in silicon gel and investigated potential applications in robotics in space and engineering industry, such as tunneling, adaptation on diverse terrain.
- Designed and modified mold CAD model for study- and testing-purpose.









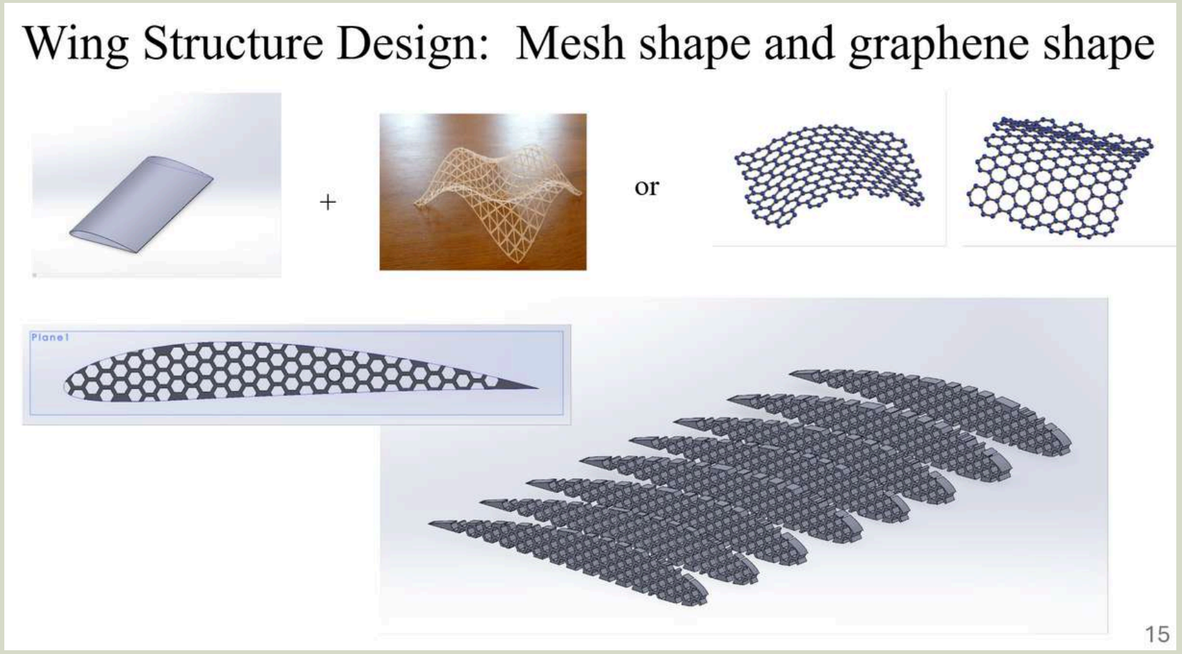
Project 3D Printing: 3D-Model Design

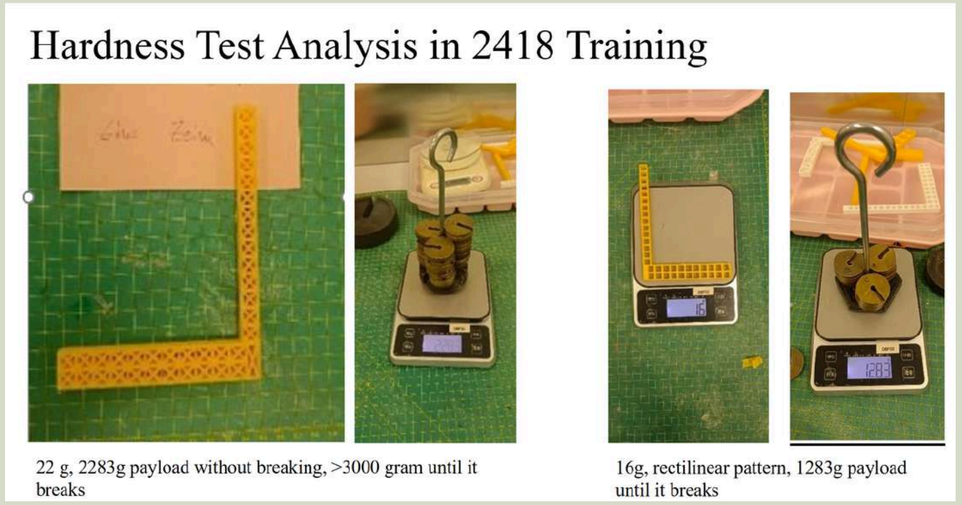
Oct - Nov, 2024

School Project: Design, Build and Fly Team-Aircraft Design

- A material study and testing on Weight reduction design, Structure Analysis on ribs using 3D print filament material for testing
- Hardness test study

Relevant courses:
Mech2419: Properties of Materials: B+
MechH2418: Training (P/F): Passed
Mech3409: Mechanics of Solid: A





MECH 2418	Engineering training	2023-24 Sum Sem	P	6.00	

3D Material & 3D-Model Design

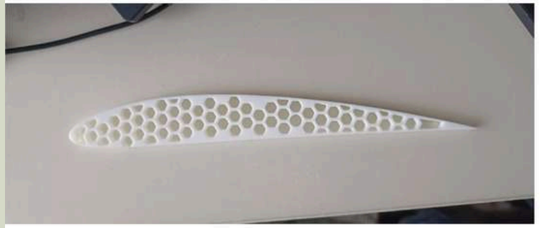
3D Print Study: Weight Reduction

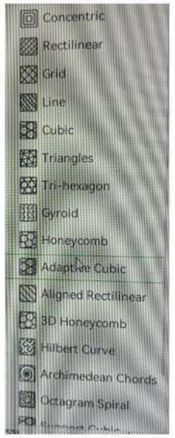
Oct - Nov, 2024

3D Printing Filament Used: Bambu PLA Basic

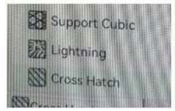
Mass: 10.09g
Wing span: ~20cm
Infill pattern: Grid

Mass: 10.09g
Wing span: ~20cm
Infill density: 10%





Infill pattern



5

Comparison Table on Weight of Different Materials

Type of Material	Weight (gram)
Basic PLA	~7.7 grams
Generic PLA (PPA)	~7.3 grams
Balsa Wood	2.34 grams
ASA Aero	~4.46gram (grid pattern), <4 gram if (gyroid pattern)
PolyAir	Not yet tested
PolyLight	Not yet tested
SLA	2.5gram (1:2 scale), ~20 gram (1:1 scale)

Result Of Material Testing: PPA vs PLA

Comparison on current existed PLA filament:
PPA(Generic PLA) VS Basic PLA

Left one is PPA (~7.3g) and the right one is basic PLA (~7.7g)



Balsa Wood



Mass properties of Part2
Configuration: Default
Coordinate system: -- default --

Density = 0.00 grams per cubic millimeter

Mass = 2.34 grams

Volume = 14623.70 cubic millimeters

Surface area = 15174.91 square millimeters

Center of mass: (millimeters)
X = 78.72
Y = 6.46
Z = 5.00

Principal axes of inertia and principal moments of inertia: (grams * square millimeters)
Taken at the center of mass.
Ix = (1.00, 0.01, 0.00) Px = 149.41
Iy = (-0.01, 1.00, 0.00) Py = 5541.71
Iz = (0.00, 0.00, 1.00) Pz = 5652.12

Moments of inertia: (grams * square millimeters)
Taken at the center of mass and aligned with the output coordinate system. (Using positiv
Lxx = 149.78 Lxy = 44.57 Lxz = -0.03
Lyx = 44.57 Lyy = 5541.35 Lyz = 0.00
Lzx = -0.03 Lzy = 0.00 Lzz = 5652.12

Moments of inertia: (grams * square millimeters)
Taken at the output coordinate system. (Using positive tensor notation.)
Ixx = 305.87 Ixy = 1234.09 Ixz = 920.88
Iyx = 1234.09 Iyy = 20097.79 Iyz = 75.56
Izx = 920.88 Izy = 75.56 Izz = 20247.67

New Material Choice

- ASA Aero: Self-foaming material, Expands when heating, so the density is low
- ASA Aero- Bambu Lab

Product Features

- Tough with Good Impact Resistance
- Ultra-lightweight & Low Density
- Water & UV Resistance
- Comes with High Temperature Reusable Spool
- Diameter: 1.75 mm +/- 0.03 mm

Cautions for Use

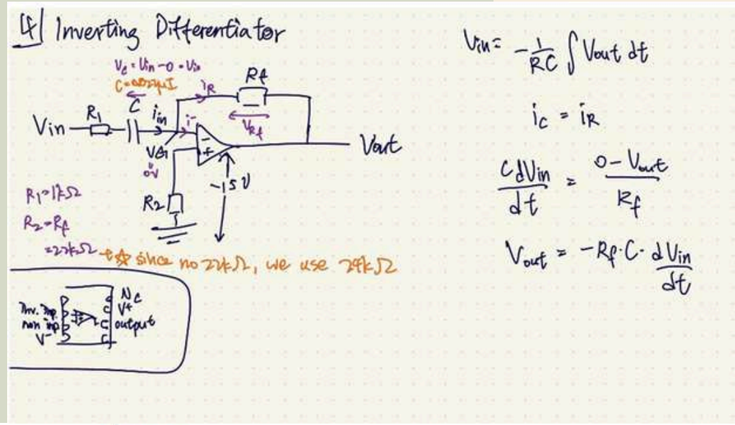
- Dry out before Use
- AMS Compatible
- Enclosure Printers Required

	Bambu ASA Aero	Bambu PLA Aero	Wood PLA
Toughness (Impact Strength - XY)	**32.0 kJ/m²**	24.5 kJ/m²	7.9 kJ/m²
Layer Adhesion (Impact Strength - Z, Dry State)	**3.4 kJ/m²**	2.5 kJ/m²	1.1 kJ/m²
Layer Strength (Bending Strength - Z)	**24 MPa**	15 MPa	15 MPa

School Training: Basic Electronics

- Training module includes:
- Understanding the usage of LabVIEW, data acquisition board to perform different virtual simulations, such as temperature input, frequency, pneumatic and LED control using relay
- Understanding the usage of operational amplifier, with different logic gate design

2024 Summer



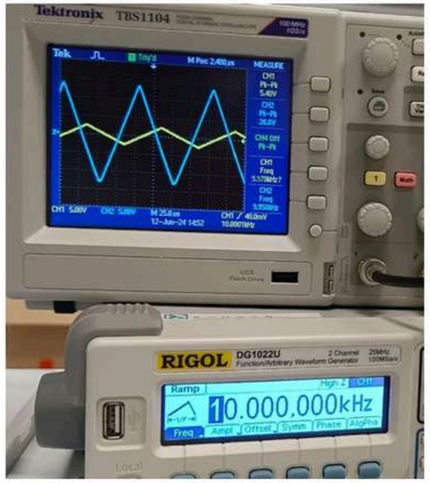


Figure 171 Waveform when frequency is 10kHz

Department of Mechanical Engineering, HKU — Report for MECH2418 Engineering Training

Inverting differentiator

In this exercise, we connect the capacitor with the resistor in series. This connection method allows the capacitor to perform integration of the input signal which produced a integrated waveform as shown in calculation at the VOUTPUT. The resistor for earthing is changed from 22k Ohms to 24k Ohms.

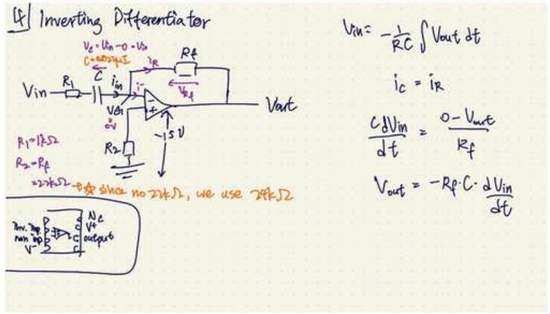


Figure 167 Circuit diagram of inverting differentiator

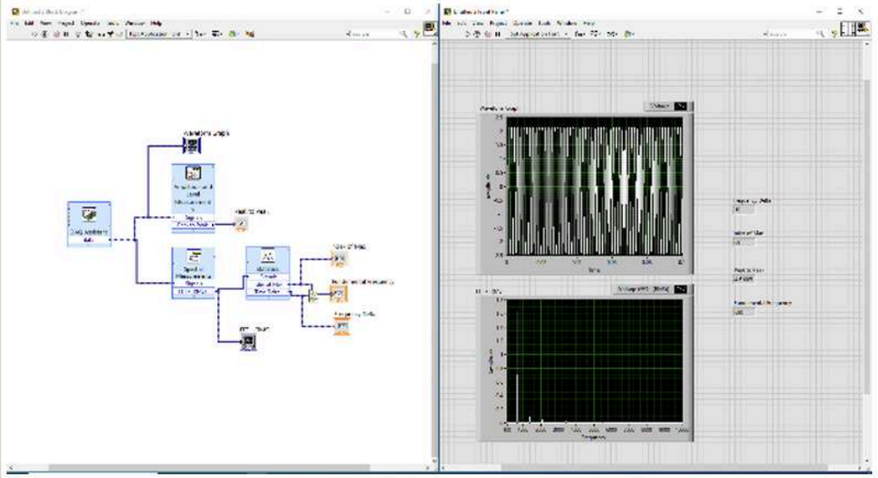


Figure 12 Ladder diagram for measurement for frequency of 0.1 – 10kHz

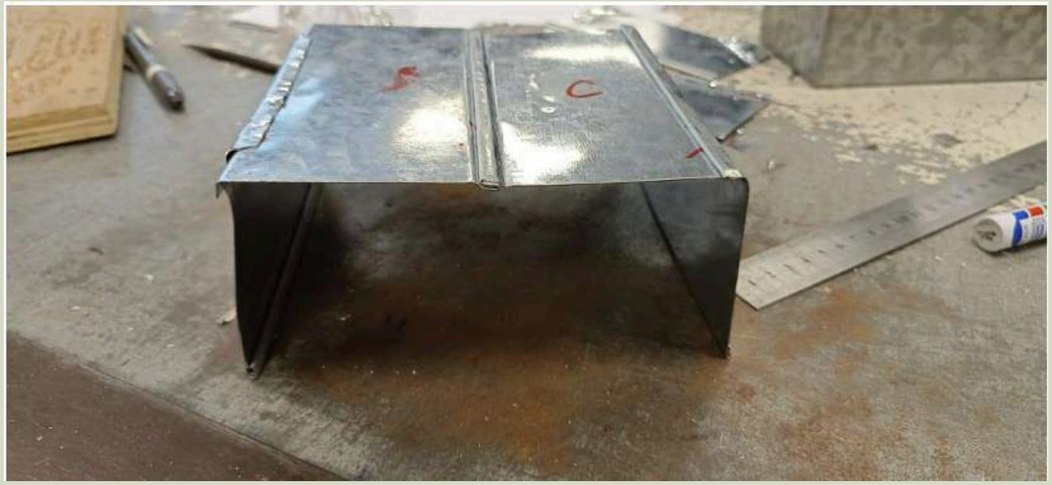
School Training: Workbench practice

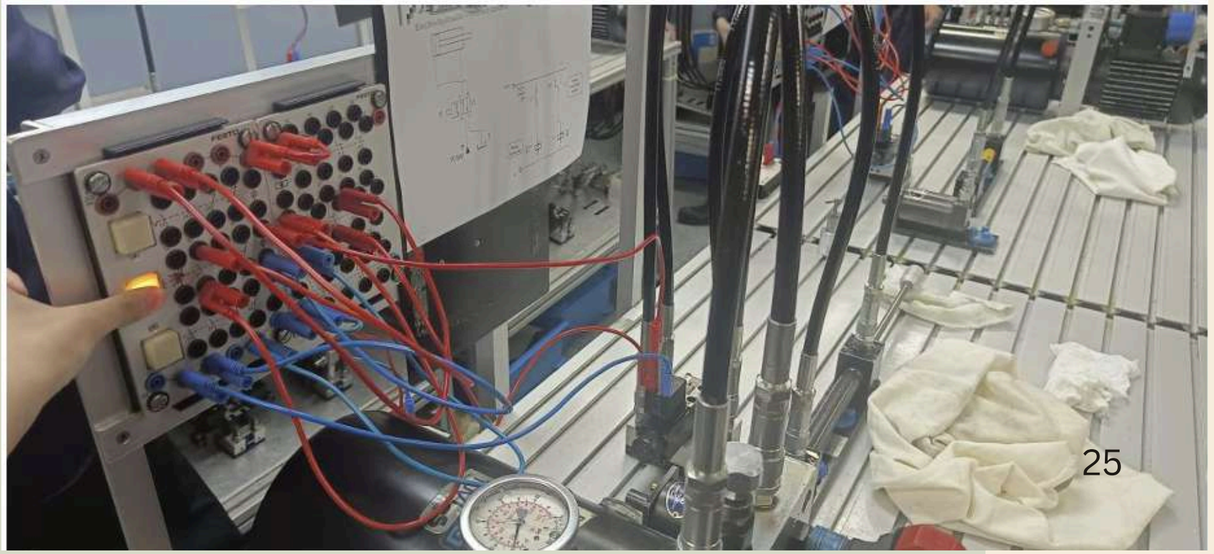
2024 Summer

- Training module includes:
- Hand-crafted a metal cover and also a metal box using mechanical tools, including hammer, metal sheet-cutting machine,
- Experience in making a hydraulic pump using circuits for electrical connection
- Experience in tapping-holes making using drilling machine









School Training: Workbench Practice

2024 Summer

- 3-person project of manufacturing a rack using workbench tools and aluminium metals
- Grinding machine practice to polish the surface of an aluminum block
- Issue appeared during the tapping process of my teammate; therefore, the screw was not able to fit into my part, which is the shaft part of the vise

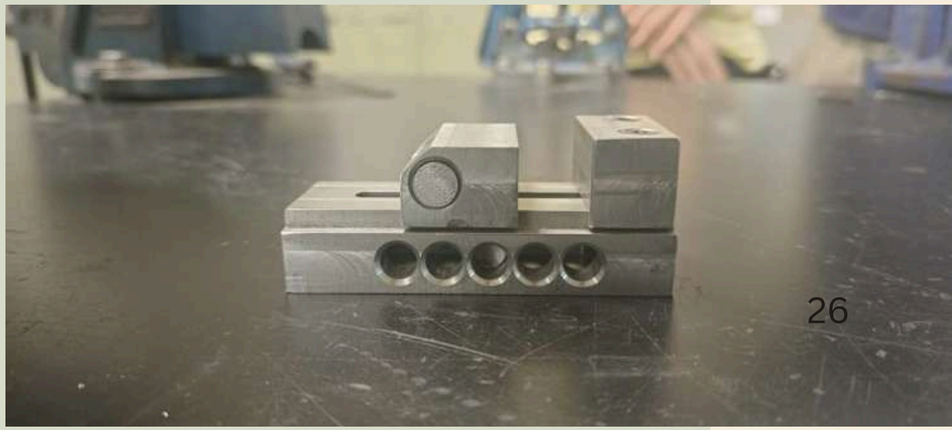












School Training: CAD/CAM

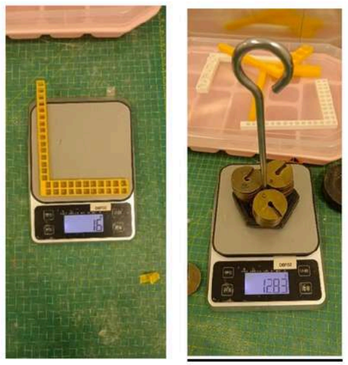
2024 Summer

- Design a 2 version of L-beam structure with biggest loading
- Result: the one with crossed-section rhombus structure is able to withstand >3000 g payload, while the one with rectangular cross section rhombus structure was able to withstand 1283g payload

Hardness Test Analysis in 2418 Training

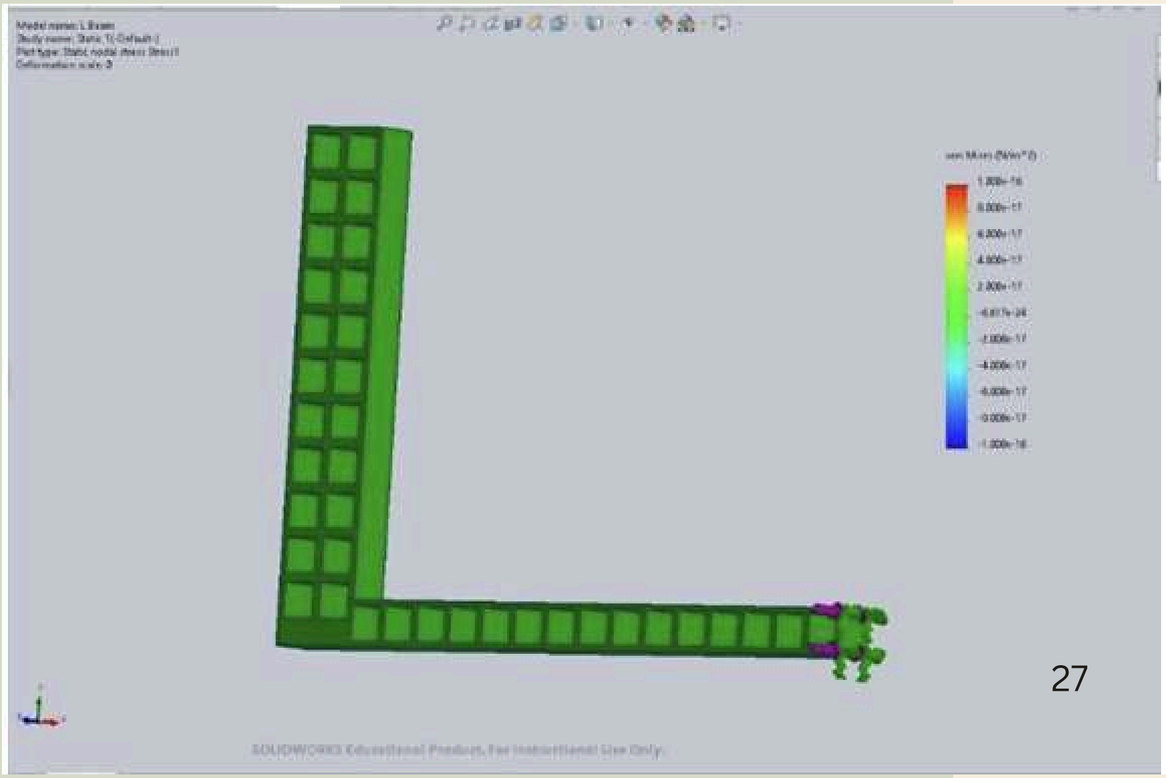


22 g, 2283g payload without breaking, >3000 gram until it breaks



16g, rectilinear pattern, 1283g payload until it breaks





School Training: CAD/CAM

2024 Summer

- Learnt fundamentals of G-code and CNC milling
- Draw a building layout of a room in HKU on AutoCAD
- Learnt fundamentals of kinematic analysis on SW

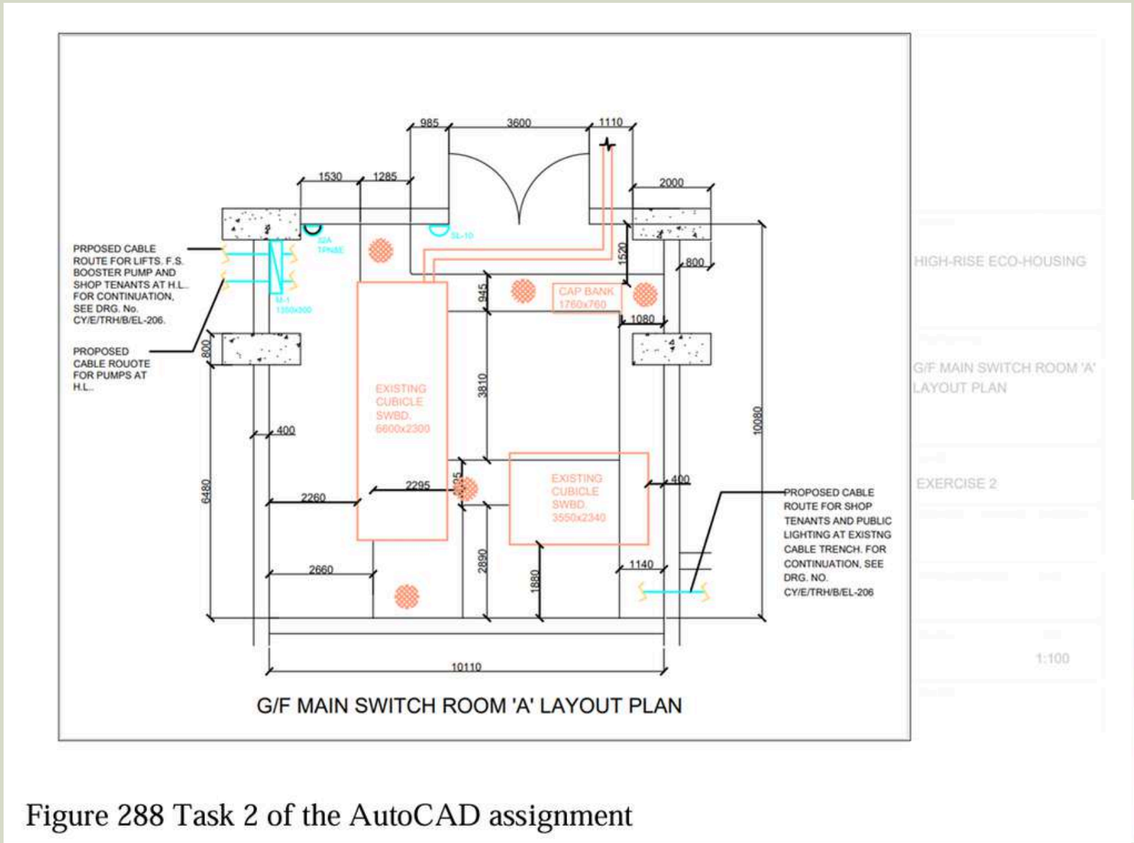


Figure 288 Task 2 of the AutoCAD assignment

Practice

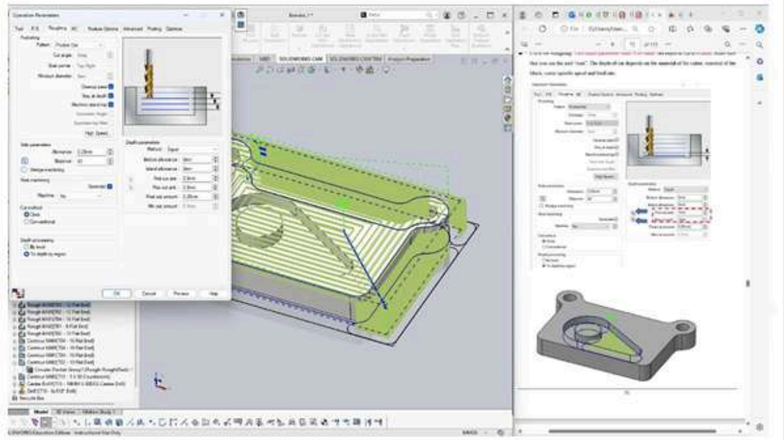
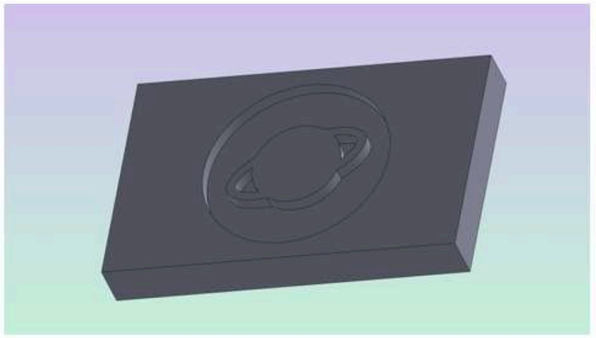
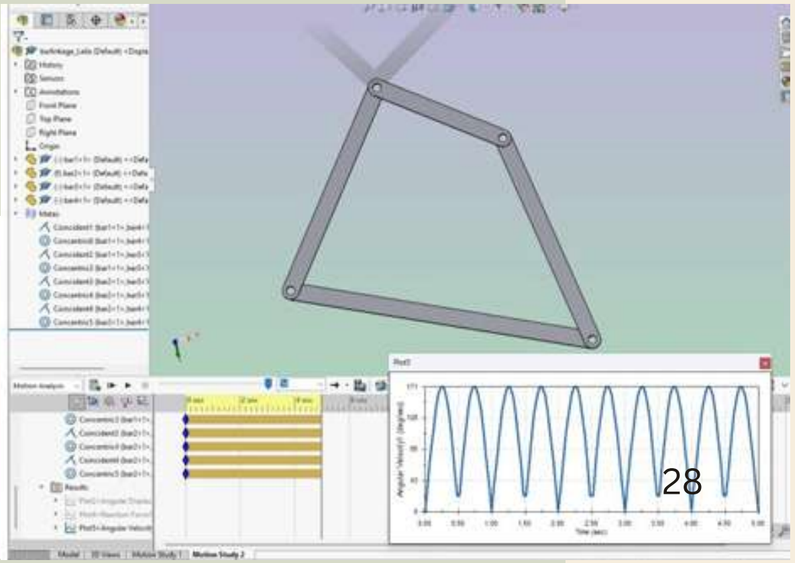


Figure 283





Advanced Mechanical Engineering Core Course Performance

2024 - 2026

4) Discipline Core Courses - Advanced Courses (42 credits)

36.0 Credits Added

IP ----> 6.0 Credits

2401	MECH 3407	6.0	B-	Advanced partial differential
2401	MECH 3409	6.0	A	Mechanics of solids
2401	MECH 3418	6.0	A-	Dynamics and control
2402	MECH 3402	6.0	A-	Engineering thermodynamics
2402	MECH 3408	6.0	B-	Mechanics of fluids
2402	MECH 3427	6.0	A+	Design and Manufacture
2502	MECH 4410	6.0	IP	Engineering & technology mana

LATX 7516	Aircraft stress analysis and f	6.00	Grading Basis 4.3	**A**	24.00
MECH 4429	Integrated capstone experience	12.00	Grading Basis 4.3	**A-**	44.40

GPA for advanced courses: 3.52/4.3

A- grade in Capstone project and A grade in aircraft stress analysis and finite element master course (MSc)

Academic-Educational Project:

2024 - now

- Team leader of student team to plan, organize and hold academic workshops to stimulate students' interest in aviation and aerospace field
- Introduce SolidWorks, Aircraft Design, and hands-on skills for RC model aircraft manufacturing
- Lead lower-year students to join several aviation/aerospace related competitions
- Experiences in Canva, PowerPoint, Team management, Event planning











Thank you!

Project Porfolio

Zhao Zehui (Leila)
Mechanical Engineering (Year 4)
The University of Hong Kong

